
Hodge Theory

— *EYNTKA* Series —

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0.0.1 Inspiration heavily inspired by [1]
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1

Kähler Manifolds

Some facts to eventually include:

1. In real manifolds, except in dimension 4 manifolds have a unique real structure. Usually there are infinitely many (often uncountably infinitely many) complex structures on a topological manifold. Think of elliptic curves. This inspires Teichmüller theory and creates interesting moduli spaces and is the source of some of the easiest example of stacks

1.1 Basics of Complex Manifolds

Let $U \subseteq \mathbb{C}^n$. Then $f : U \rightarrow \mathbb{C}$ be holomorphic if it is holomorphic in each variable, and $F = (f_1, \dots, f_n) : U \rightarrow \mathbb{C}^n$ is holomorphic if it is also holomorphic in each component. By identifying $\mathbb{C}^n \cong \mathbb{R}^{2n}$, let $z_j = x_j + iy_j$ and $f_k = u_k + iv_k$ for $1 \leq j, k \leq n$. Certainly u_k, v_k are smooth in $x_1, y_1, \dots, x_n, y_n$ and satisfy the Cauchy Riemann equations:

$$\frac{\partial u_k}{\partial x_j} = \frac{\partial v_k}{\partial y_j} \quad \text{and} \quad \frac{\partial u_k}{\partial y_j} = -\frac{\partial v_k}{\partial x_j}$$

Conversely, given a smooth map $(u_1, v_1, \dots, u_n, v_n) : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ where each pair u_i, v_i satisfies the Cauchy-Riemann equations, then the map $(u_1 + iv_1, \dots, u_n + iv_n)$ is holomorphic. In terms of DF , given that the tangent space $T_p(\mathbb{R}^{2n})$ is written in coordinates:

$$\left\{ \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right\}$$

and the $T_{F(p)}(\mathbb{R}^{2n})$ is written in basis:

$$\left\{ \frac{\partial}{\partial u_1}, \dots, \frac{\partial}{\partial u_n}, \frac{\partial}{\partial v_1}, \dots, \frac{\partial}{\partial v_n} \right\}$$

Then we get the familiar:

$$DF(p) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \quad (1.1)$$

where A, B are the matrices for the $\frac{\partial}{\partial x_j}$ and $\frac{\partial}{\partial y_j}$ respectively; let us call this matrix *complex multiplication representation* or *CR matrix* (for Cauchy-Riemann, as it represents the relationship). This can be verified computationally (without needing to find a nice basis) by checking whether the such a matrix commutes with:

$$J := \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$

Now upgrading this to a manifold:

Definition 1.1.1: Complex Manifold

A topological manifold M is a *complex manifold* if there exists a *complex atlas* or *complex structure* $(U_\alpha, \varphi_\alpha)$ where:

1. $M = \bigcup_\alpha U_\alpha$ is an open covering
2. $\varphi_\alpha : U_\alpha \rightarrow V_\alpha \subseteq \mathbb{C}^n$ is a homeomorphism (where by [4] n is independent of α)
3. If $U_\alpha \cap U_\beta \neq \emptyset$ then

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is a biholomorphism.

If M, N are complex manifolds, $f : M \rightarrow N$ is a holomorphic map if it is holomorphic in coordinates. The map f is biholomorphic if it has a holomorphic inverse.

Given open $U \subseteq M$, let $\mathcal{O}(U) = \{f : U \rightarrow \mathbb{C} \mid f \text{ holomorphic}\}$ and similarly $\mathcal{O}^*(U)$ be the set of nowhere-vanishing (nowhere-zero) holomorphic functions.

Example 1.1.2: Complex Manifolds

The details that any of these are complex manifolds follows almost entirely from differential geometry, see [4].

1. $\mathbb{C}^n$ and every open subset naturally is a complex manifold
2. A special open subset is $\text{GL}(n, \mathbb{C})$.
3. Complex projective space $\mathcal{P}^n$ is a compact complex manifold
4. the complex Grassmannian $G(k, n)$ of k -dimensional subspaces of $\mathbb{C}^n$ are compact complex manifolds. Note that $G(k, n)$ can be viewed as a quotient of $V(k, n) = \{W \in M_{n \times k}(\mathbb{C}) \mid \text{rank}(W) = k\}$ by:

$$G(k, n) := \frac{V(k, n)}{\text{GL}(k, \mathbb{C})}$$

where the multiplication is on the right.

5. Complex Lie groups

6. (complex) varieties. Elliptic curves being a source of many examples.
7. If M is not orientable, it cannot be a complex manifold (see [ref:HERE](#)).

Proposition 1.1.3: Maximum Principle On Complex Manifolds

let M be a compact, connected, complex manifold and $f : M \rightarrow \mathbb{C}$ holomorphic. Then f is constant

Proof:

Use the fact that the maximum modulus principle holds for holomorphic functions of several variables and analytic continuation (see [5])

Let $F : M \rightarrow N$ be a holomorphic map and choose coordinates so that $\tilde{F} : U \rightarrow V$ where $U \subseteq \mathbb{C}^n$, $V \subseteq \mathbb{C}^n$; let $\tilde{F} = (f_1, \dots, f_m)$ in coordinates. Then:

$$D\tilde{F}(p) : U \rightarrow V \quad D\tilde{F}(p)(v) = \left(\frac{\partial f_i}{\partial z_j}(p) \right) \cdot v$$

By the Cauchy-Riemann equations, the matrix representation is of the form in equation (1.1). It is not hard to see that the inverse respects this form, recovering the complex inverse function theorem for manifolds. Naturally, this also recovers the implicit version, and more generally the Rank theorem, see [4]. Given we have the rank theorem, we can recover similar results for submanifolds. Let $F : M \rightarrow N$ be a holomorphic map. Then:

$$\text{rank}_p(F) = \text{rank}(D(\psi \circ F \circ \varphi^{-1})(\varphi(p)))$$

for a choice of local coordinates¹. Then:

Theorem 1.1.4: Complex Submanifold

Let $F : M \rightarrow N$ be a holomorphic map, $q \in F(M)$, $X = F^{-1}(\{q\})$. Suppose $\text{rank}_x(F) = k$ for all $x \in U$ where U contains X . Then X is a complex manifold and:

$$\text{codim} X = k$$

Proof:

exactly the same as in [4]

There is a *pivotal* difference between the existence of real and complex submanifolds

Proposition 1.1.5: All compact complex manifolds of $\mathbb{C}^n$ are trivial

There are no compact complex submanifold of $\mathbb{C}^n$ of dimension greater than 0

¹the rank is independent of this choice

Proof:

Let $M \subseteq \mathbb{C}^n$ be a submanifold. Then each coordinate function z_i restricts to a holomorphic function on M . But M is compact, hence by proposition 1.1.3 z_i must be locally constant, hence $\dim M = 0$.

Thus, there cannot be a Whitney Embedding theorem for holomorphic manifolds. We shall prove in [ref:HERE](#) that there is the *Kodaira Embedding Theorem* which gives the conditions for a compact complex manifold to be embedded in $\mathbb{P}^n$.

To define more complicated structure on a complex manifold, we need to have the appropriate complex vector bundle theory:

Definition 1.1.6: Complex Vector Bundle

A *holomorphic vector bundle* E over a complex manifold M is a complex manifold E together with a holomorphic map $\pi : E \rightarrow M$ such that:

1. For each $x \in M$, the fiber $E_x = \pi^{-1}(x)$ is a complex vector space of dimension d (the rank of E).
2. There exists an open covering $\{U_\alpha\}$ of M and biholomorphic maps

$$\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{C}^d$$

such that

- (a) $p_1(\Phi_\alpha(x)) = x$ for all $x \in U$, where $p_1 : U_\alpha \times \mathbb{C}^d \rightarrow U_\alpha$ denotes projection on the first factor, and
- (b) For every $x \in U_\alpha$ the map $p_2 \circ \Phi_\alpha|_{E_x} : E_x \rightarrow \mathbb{C}^d$ is an isomorphism of complex vector spaces.

Call E the total space of the bundle and M its base. The covering $\{U_\alpha\}$ is called a local trivializing cover of M and the biholomorphisms $\{\Phi_\alpha\}$ local trivializations. When $d = 1$ we often refer to E as a line bundle.

If $U \subseteq M$, then a holomorphic map $\sigma : U \rightarrow E$ such that $\pi \circ \sigma = \text{id}|_U$ is called a *holomorphic section* of M . The collection of all section of E of an open set U is denoted $\Gamma(U, E)$ and these are each a $\mathcal{O}(U)$ -modules.

A holomorphic vector bundle can also be described using transition functions, namely, given a covering of M by open sets U_α take holomorphic maps:

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}(d, \mathbb{C})$$

such that

$$g_{\alpha\beta} \cdot g_{\beta\gamma} = g_{\alpha\gamma}$$

on $U_\alpha \cap U_\beta \cap U_\gamma$. The maps $g_{\alpha\beta}$ are defined by the commutative diagram:

$$\begin{array}{ccc} & \pi^{-1}(U_\alpha \cap U_\beta) & \\ \Phi_\beta \swarrow & & \searrow \Phi_\alpha \\ (U_\alpha \cap U_\beta) \times \mathbb{C}^d & \xrightarrow{(\text{id}, g_{\alpha\beta})} & (U_\alpha \cap U_\beta) \times \mathbb{C}^d \end{array}$$

A holomorphic line bundle over M is thus given by a collection $\{U_\alpha, g_{\alpha\beta}\}$, where U_α is an open cover of M and the $\{g_{\alpha\beta}\}$ are *nowhere-zero* holomorphic functions defined on $U_\alpha \cap U_\beta$, i.e. $g_{\alpha\beta} \in \mathcal{O}^*(U_\alpha \cap U_\beta)$ satisfying the cocycle condition.

If $F : N \rightarrow M$ is a holomorphic map and $\pi : E \rightarrow M$ a holomorphic vector bundle, then we can define a pullback:

$$h_{\alpha\beta} : F^{-1}(U_\alpha) \cap F^{-1}(U_\beta) \rightarrow \text{GL}(d, \mathbb{C}) \quad h_{\alpha\beta} = g_{\alpha\beta} \circ F$$

for local trivialization given by U_α 's and transition functions $g_{\alpha\beta}$. This gives a diagram:

$$\begin{array}{ccc} F^*(E) & \xrightarrow{\bar{F}} & E \\ \pi^* \downarrow & & \downarrow \pi \\ N & \xrightarrow{F} & M \end{array}$$

If we look at line bundles in particular, as their tensor product preserves dimension (since $1 \cdot 1 = 1$), they are of particular interest. If L, L' are two line bundle cover X subject to the same local trivialization with transition functions $g_{\alpha\beta}, g'_{\alpha\beta}$, then:

$$h_{\alpha\beta} = g_{\alpha\beta} g'_{\alpha\beta}$$

define a line bundle, and is denoted $L \otimes L'$. Similarly, the tensor product can be defined in the usual way and the transition function would be the above multiplication. The dual bundle of L , denoted L^{-1}, L^* or $\text{hom}(L, \mathbb{C})$ is defined as

$$h_{\alpha\beta} = (g_{\alpha\beta})^{-1}$$

Then $L \otimes L^{-1}$ is the trivial bundle.

Example 1.1.7: Complex Vector Bundle

For now I'm relegating all examples to [3]. An important one to point from algebraic geometry is that the *tautological bundle* on $\mathbb{P}^n$ is denoted $\mathcal{O}(-1)$, and its dual $\mathcal{O}(1)$. By some work in algebraic geometry, it is easy to verify that this notation is justified as it corresponds to a hyperplane $H \cong \mathbb{P}^{n-1} \subseteq \mathbb{P}^n$. Note this hyperplane can be given as the zero-locus of a section of a line bundle.

A similar result can be done for $G(n, k)$.

Next, let $L \rightarrow M$ be a line bundle with local trivialization $\{U_\alpha, \varphi_\alpha\}_\alpha$. Then a section $\sigma : M \rightarrow L$ may be given by taking a collection of holomorphic function $f_\alpha \in \mathcal{O}(U_\alpha)$ and setting:

$$\sigma(x) = f_\alpha(x) \varphi_\alpha^{-1}(x, 1)$$

Thus, for any $x \in U_\alpha \cap U_\beta$:

$$f_\alpha(x) = g_{\alpha\beta}(x) \cdot f_\beta(x)$$

showing that line bundles are functions “up to” a transition factor on each intersection. We shall see in [ref:HERE](#) that all line bundles on $\mathbb{P}^n$ will be of the form $\mathcal{O}(d)$ that is homogeneous degree n polynomials.

1.1.1 Almost Complex Structure

A natural question to ask is if we may impose a complex structure per tangent space that are compatible and see if this produces a complex manifold. Let us first show that such a map always exists for a complex manifold. Let $(U_\alpha, \varphi_\alpha)$ be coordinate charts for a complex manifold M . Then any change of coordinate maps are holomorphic, so the matrix representation of $D(\varphi_\beta \circ \varphi_\alpha^{-1})$ must represent complex multiplication. In Thus, there is a natural map:

$$J_p : T_p(M) \rightarrow T_p(M)$$

where per coordinate:

$$J \left(\frac{\partial}{\partial x_j} \right) = \frac{\partial}{\partial y_j} \quad \text{and} \quad J \left(\frac{\partial}{\partial y_j} \right) = -\frac{\partial}{\partial x_j}$$

Note that J is a smooth $(1,1)$ tensor. Recall a $(1,1)$ tensor is point-wise:

$$F_p : T_p^*(M) \times T_p(M) \rightarrow \mathbb{R}$$

and by currying we transform $F(f, v)$ to a function between tangent spaces. Note that:

$$J^2 = -1$$

Hence J captures the complex structure at each point. If an arbitrayr smooth manifold M admits a $(1,1)$ smooth tensor J such that $J^2 = -1$, we give it a name:

Definition 1.1.8: Almost Complex Manifold

Let M be a smooth manifold. If there exists a smooth map:

$$J : TM \rightarrow TM$$

such that $J^2 = -1$, then (M, J) is called an *almost complex structure*.

Given (M, J) then J_p implies a complex structure on $T_p M$, and hence M must be even-dimensional. It is also a good exercise to show that (M, J) is orientable with an orientation imposed by J . In the special case where M is an oriented two-dimensional manifold with Riemann metric $\langle -, - \rangle$, then we can define J pointwise by choosing an oriented orthonormal basis at each tangent space (note that this is *not* a frame, ex. S^2 does not have a non-vanishing frame) and letting:

$$J_p(v_1) = v_2 \quad J_p(v_2) = -v_1$$

Show that this is independent of choice of orthonormal oriented basis. In particular, if there is another Riemann metric $\langle -, - \rangle'$ conformally equivalent to $\langle -, - \rangle$, then the two matrices define the same almost complex structure.

The term “almost complex” is because this is not enough to define a complex structure. For now, we simply state the necessary condition.

Theorem 1.1.9: Newlander-Nirenberg Theorem

Let (M, J) be an almost complex manifold and N a $(1, 2)$ tensor given by:

$$N(X, Y) = [JX, JY] - [X, Y] - J[X, JY] - J[JX, Y]$$

It is a computation to see that $N(X, Y) = -N(Y, X)$ and N is called the *torsion* of J .

Then (M, J) is a *complex manifold* inducing the almost-complex structure J if and only if $N(X, Y) = 0$ for all vector fields X, Y on M . If J induces a complex structure, then it is said to be *integrable*.

Proof:

[2] quoted a result on p.16

In fact, with some dimension arguments, show that in this case

$$N(X, Y) = 0$$

showing that the obstruction happens only in higher (even) dimensions.

Thus, it is not enough to have smoothly varying complex structure at each point. (Cool remarks about the progress of classification related to parallelizable spheres at the end of section 1.2.1)

1.1.2 Complex Differential Forms

Next, let us consider differential forms on almost complex and complex manifolds. If (M, J) is an almost complex manifold, then the (real) tangent space $T_p M$ has the following decomposition after complexification:

$$T_{p, \mathbb{C}} M = T_p M \otimes \mathbb{C} = T'_p M \oplus T''_p M$$

where $T''_p M = \overline{T'_p M}$ and $T'_p M$ is the i -eigenspace of J_p acting on $T_{p, \mathbb{C}} M$. Given J_p , we get a complex vector-space isomorphism:

$$T_p M \rightarrow T'_p M \quad v \mapsto v - iJ_p(v)$$

If J is integrable, then there are holomorphic coordinates $\{z_1, \dots, z_n\}$, there are induced local coordinate frame with the isomorphism:

$$\frac{\partial}{\partial x_j} \mapsto \frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j}$$

From these, we can define the Wirtinger derivatives:

$$\frac{\partial}{\partial z_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right) \quad \text{and} \quad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right)$$

The vectors $\frac{\partial}{\partial z_j}$ form a basis for $T'_p M$. Recall from complex that $\frac{\partial}{\partial \bar{z}_j}$ can be used to determine if a function is holomorphic, namely if $f : U \rightarrow \mathbb{C}$ where U has local coordinates, then f is holomorphic if:

$$\frac{\partial}{\partial \bar{z}_j}(f) = 0 \quad \forall j$$

Recall to that $\frac{\partial}{\partial z_j}(f)$ coincides with the partial derivatives of f with respect to z_j , justifying the notation.

Definition 1.1.10: Holomorphic Tangent Space

Let M be a complex manifold. Then $T_p^h M$ is called the *holomorphic tangent space* at p , and $T^h M$ the *holomorphic tangent bundle*.

Note that if there are two set of complex coordinates $\{z_1, \dots, z_n\}$ and $\{w_1, \dots, w_n\}$, then the change of basis matrix is given by a matrix of holomorphic function $\left(\frac{\partial w_k}{\partial z_j}\right)_{kj}$ showing that $T^h M$ is indeed a holomorphic bundle.

We now get the following crucial example:

Proposition 1.1.11: Tangent Bundle of Projective Space

let $\mathbb{P}^n$ be the complex projective space. Then:

$$T\mathbb{P}^n \simeq \text{Hom}(\mathcal{T}, E/\mathcal{T})$$

where $E = \mathbb{P}^n \times \mathbb{C}^{n+1}$ is the trivial $n+1$ -rank complex bundle and $\mathcal{T} = \mathcal{O}(-1)$ is the tautological bundle. The holomorphic tangent bundle $T^h \mathbb{P}^n$ can be identified with:

$$\text{Hom}_{\mathbb{C}}(\mathcal{T}, E/\mathcal{T})$$

Proof:

Let $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}^n$ be the natural projection. Given $\lambda \in \mathbb{C}^*$, let M_λ denote multiplication by λ in $\mathbb{C}^{n+1} \setminus \{0\}$. Then, for every $v \in \mathbb{C}^{n+1} \setminus \{0\}$ identify $T'_v(\mathbb{C}^{n+1} \setminus \{0\}) \cong \mathbb{C}^{n+1}$ and consider the following commutative diagram with $\mathbb{C}$ -linear maps:

$$\begin{array}{ccc} \mathbb{C}^{n+1} & \xrightarrow{M_\lambda} & \mathbb{C}^{n+1} \\ \pi_{*,v} \searrow & & \swarrow \pi_{*,\lambda v} \\ & T_{[v]}(\mathbb{P}^n) & \end{array}$$

Now, the map $\pi_{*,v} : \mathbb{C}^{n+1} \rightarrow T_{[v]}(\mathbb{P}^n)$ is surjective and its kernel is the line $L = \mathbb{C} \cdot v$. Hence we get a family of $\mathbb{C}$ -linear isomorphisms

$$\mathfrak{p}_v : \frac{\mathbb{C}^{n+1}}{L} \rightarrow T_{[v]}(\mathbb{P}^n); \quad v \in L, v \neq 0$$

where:

$$\mathfrak{p}_v = \lambda \mathfrak{p}_{\lambda v}$$

Thus, define the map:

$$\Theta : \text{Hom}_{\mathbb{C}}(\mathcal{T}, E/\mathcal{T}) \rightarrow T^h \mathbb{P}^n.$$

Let

$$\xi \in \text{Hom}_{\mathbb{C}}(\mathcal{T}, E/\mathcal{T})_{[z]} = \text{Hom}_{\mathbb{C}}(\mathcal{T}_{[z]}, (E/\mathcal{T})_{[z]}) \cong \text{Hom}_{\mathbb{C}}(L, \mathbb{C}^{n+1}/L)$$

then we set

$$\Theta(\xi) := \mathfrak{p}_v(\xi(v)) \text{ for any } v \in L, v \neq 0.$$

Note that this is well defined since

$$\mathfrak{p}_{\lambda v}(\xi(\lambda v)) = \lambda^{-1} \mathfrak{p}_v(\lambda \xi(v)) = \mathfrak{p}_v(\xi(v)).$$

Alternatively one may define

$$\Theta(\xi) = \left. \frac{d}{dt} \right|_{t=0} (\gamma(t)).$$

where $\gamma(t)$ is the holomorphic curve through $[z]$ in $\mathbb{P}^n$ defined by

$$\gamma(t) := [v + t\xi(v)].$$

One then has to show that this map is well defined. It is straightforward, though tedious, to verify that Θ is an isomorphism of vector bundles.

A similar result can be proven for the Grassmannian, namely:

$$T\text{Grass}(k, n) \cong \text{Hom}(\mathcal{U}, E/\mathcal{U}) \quad \text{and} \quad T^h\text{Grass}(k, n) \cong \text{Hom}_{\mathbb{C}}(\mathcal{U}, E/\mathbb{C})$$

where $\mathcal{U}$ is the universal bundle of $\text{Grass}(k, n)$ and $E = \text{Grass}(k, n) \times \mathbb{C}^n$.

Now, looking at the cotangent bundle and its complexification, we get the similar result:

$$T_{p,\mathbb{C}}^*(M)T_p^{1,0}M \oplus T_p^{0,1}M \quad T_p^{1,0}M = \overline{T_p^{0,1}M}$$

Given local coordinates $\{z_1, \dots, z_n\}$ on U where $z_j = x_j + iy_j$, the one-forms $dz_j = dx_j + idy_j$ and $d\bar{z}_j = dx_j - idy_j$ are the dual coframe to

$$\left\{ \frac{\partial}{\partial z_1}, \dots, \frac{\partial}{\partial z_n} \right\} \quad \text{and} \quad \left\{ \frac{\partial}{\partial \bar{z}_1}, \dots, \frac{\partial}{\partial \bar{z}_n} \right\}$$

and so $dz_1, \dots, dz_n$ are a holomorphic frame for the holomorphic bundle $T^{1,0}M$.

By elementary differential geometry, we can get the decomposition:

$$\bigwedge^k (T_{p,\mathbb{C}}^*M) = \bigoplus_{a+b=k} T_p^{a,b}(M)$$

where:

$$T_p^{a,b}(M) = \underbrace{T_p^{1,0}(M) \wedge \dots \wedge T_p^{1,0}(M)}_{a \text{ times}} \wedge \underbrace{T_p^{0,1}(M) \wedge \dots \wedge T_p^{0,1}(M)}_{b \text{ times}}$$

Thus:

$$\bigwedge^k (T_{\mathbb{C}}^*M) = \bigoplus_{a+b=k} T^{a,b}(M)$$

Definition 1.1.12: Wedge Sections

Let $\mathcal{A}(U)$ (resp. $\mathcal{A}^{a,b}(U)$) be the $C^\infty(U)$ -module of local sections of the bundle $\bigwedge^k (T_{\mathbb{C}}^*M)$ (resp. $\bigoplus_{a+b=k} T^{a,b}(M)$).

Certainly:

$$\mathcal{A}(U) = \bigoplus_{a+b=k} \mathcal{A}^{a,b}(U)$$

For future reference, the local frame for $T^{a,b}$ is given by:

$$dz_I \wedge d\bar{z}_J = dz_{i_1} \wedge \cdots \wedge dz_{i_a} \wedge d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_b}$$

where I, J run over strictly increasing index sets:

$$1 \leq i_1 \leq \cdots \leq i_a \leq n \quad \text{and} \quad 1 \leq j_1 \leq \cdots \leq j_b \leq n$$

A special case to distinguish is $T^{k,0}M$ which are the holomorphic bundles of rank $\binom{n}{k}$. To emphasize these are holomorphic bundles, these shall be denoted:

$$\Omega_M^k = T^{k,0}M$$

As before, $\Omega^k(U)$ is the $\mathcal{O}(U)$ -module of holomorphic sections.

Definition 1.1.13: Canonical Bundle

A special case is Ω_M^n which is called the *canonical bundle* and is denoted K_M for historical purposes.

Example 1.1.14: Canonical Bundle On Projective Space

Let $M = \mathbb{P}^1$. Cover M by the usual two coordinate neighborhoods (U_0, φ_0) , (U_1, φ_1) where $\varphi_1 \circ \varphi_0^{-1} : \mathbb{C}^* \rightarrow \mathbb{C}^*$ is given by:

$$w = \varphi_1 \circ \varphi_0^{-1}(z) = \varphi_1([(1, z)]) = 1/z.$$

This means that the local sections dz , dw of the holomorphic cotangent bundle are related by

$$dz = -(1/w)^2 dw.$$

It follows that $g_{01}[(z_0, z_1)] = -(z_0/z_1)^2$. Hence $K_{\mathbb{P}^1} \cong \mathcal{O}(-2) = \mathcal{T}^2$. More generally:

$$K_{\mathbb{P}^n} \cong \mathcal{O}(-n-1) = \mathcal{T}^{n+1}$$

Now, given $\mathcal{A}^k(U)$ where $U \subseteq M$ is open and M is *smooth*, there is a unique operator called the *exterior derivative*:

$$d : \mathcal{A}^k(U) \rightarrow \mathcal{A}^{k+1}(U) \quad k \geq 0$$

where

1. d is $\mathbb{C}$ -linear
2. For $f \in \mathcal{A}^0(U) = C^\infty(U)$, df is the 1-form on U which acts on vector fields as:

$$df(X) = X(f)$$

3. Given $\alpha \in \mathcal{A}^r(U)$ and $\beta \in \mathcal{A}^s(U)$, then:

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^r \alpha \wedge d\beta$$

4. $d \circ d = 0$

This defines a map $C^\infty(U) \rightarrow \mathcal{A}^1(U)$, namely given a local frame $\{X_1, \dots, X_m\}$ on open $U \subseteq M$ and dual coframe $\{\xi_1, \dots, \xi_m\}$, then:

$$df = \sum_{i=1}^m X_i(f) \xi_i$$

and given a choice of local coordinates $\{z_1, \dots, z_n\}$ for U :

$$df = \sum_{j=1}^n \frac{\partial f}{\partial x_j} dx_j + \sum_{j=1}^n \frac{\partial f}{\partial y_j} dy_j = \sum_{j=1}^n \frac{\partial f}{\partial z_j} dz_j + \sum_{j=1}^n \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$$

Then this chain extends by the above properties to:

$$\mathbb{C} \hookrightarrow C^\infty(U) \xrightarrow{d} \mathcal{A}^1(U) \xrightarrow{d} \mathcal{A}^2(U) \xrightarrow{d} \dots \xrightarrow{d} \mathcal{A}^{2n-1}(U) \xrightarrow{d} \mathcal{A}^{2n}(U)$$

Definition 1.1.15: Complex De Rham Cohomology

Let:

$$Z^k(U) = \ker\{d : \mathcal{A}^k(U) \rightarrow \mathbb{C}^{k+1}(U)\} \quad B^k(U) = d(\mathcal{A}^{k-1}(U))$$

where a form in $Z^k(U)$ is called *closed*, and a form in $B^k(U)$ is called *exact*. Then the *deRham cohomology group* is:

$$H_{\text{dR}}^k(U, \mathbb{C}) = \frac{Z^k(U)}{B^k(U)}$$

and the *deRham cohomology ring* is:

$$H_{\text{dR}}^*(U, \mathbb{C}) = \bigcup_{k \geq 0} H^k(U, \mathbb{C}) \quad [\alpha] \smile [\beta] = [\alpha \wedge \beta]$$

Given map $F : M \rightarrow N$, for any open $V \subseteq N$ there is a pullback:

$$F^* : \mathcal{A}^k(V) \rightarrow \mathcal{A}^k(F^{-1}(V))$$

By differential geometry, it is easy to see that F^* commutes with the exterior differential and $(G \circ F)^* = G^* \circ F^*$ hence there is a ring homomorphism:

$$F^* : H^k(V, \mathbb{C}) \rightarrow H^k(F^{-1}(V), \mathbb{C})$$

And if F is an isomorphism, so is F^* . By [4], if F is a (smooth) homotopy invariant, F^* is an isomorphism.

Now, one way d is different over complex coefficients is that we get the following splitting:

$$d(\mathcal{A}^{a,b}(U)) \subseteq \mathcal{A}^{a+1,b}(U) \oplus \mathcal{A}^{a,b+1}(U)$$

Thus, we can decompose $d = \partial + \bar{\partial}$ where ∂ and $\bar{\partial}$ are bidegree $(1, 0)$ and $(0, 1)$ respectively. Then as $d^2 = 0$:

$$\partial^2 = \bar{\partial}^2 = 0 \quad \partial \circ \bar{\partial} + \bar{\partial} \circ \partial = 0$$

where both ∂ and $\bar{\partial}$ can be shown to the Leibniz property. Importantly:

Proposition 1.1.16: Holomorphic Forms

Let M be a complex manifold, $U \subseteq M$ open. Then:

$$\ker\{\bar{\partial} : \mathcal{A}^{p,0}(U) \rightarrow \mathcal{A}^{p,1}(U)\} = \Omega^p(U)$$

That is, the kernel is the complex holomorphic p -forms on U .

Proof:

choose local coordinates $\{z_1, \dots, z_n\}$ and $\alpha \in \mathcal{A}^{p,0}(U)$. Write $\alpha = \sum_I f_I dz_I$ where I runs over the increasing index set. Then:

$$\bar{\partial}\alpha = \sum_I \sum_{j=1}^n \frac{\partial f_I}{\partial \bar{z}_j} d\bar{z}_j \wedge dz_I = 0$$

Thus

$$\frac{\partial f_I}{\partial \bar{z}_j} = 0$$

for all I and j . But then $f_I \in \mathcal{O}(U)$ for all I , hence α is a holomorphic p form.

Thus, for each p , $0 \leq p \leq n$:

$$0 \rightarrow \Omega^p(U) \hookrightarrow \mathcal{A}^{p,0}(U) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1}(U) \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \mathcal{A}^{p,n-1}(U) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,n}(U) \rightarrow 0$$

This is the Dolbeault complex, and this give:

Definition 1.1.17: Dolbeault Cohomology

$$H_{\bar{\partial}}^{p,q}(U) = \frac{\ker\{\bar{\partial} : \mathcal{A}^{p,k}(U) \rightarrow \mathcal{A}^{p,k+1}(U)\}}{\bar{\partial}(\mathcal{A}^{p,k-1}(U))}$$

When unambiguous the subscript is dropped and we simply denote it as $H^{p,q}(U)$

Proposition 1.1.18: Properties of $\bar{\partial}$ map

Let $\alpha \in \mathcal{A}^{p,q}(U)$. Then $\overline{\partial\alpha} = \bar{\partial}\bar{\alpha}$, hence α is ∂ -closed/exact if and only if $\bar{\alpha}$ is $\bar{\partial}$ -closed/exact. In particular:

$$\overline{H_{\partial}^{p,q}(U)} = H_{\bar{\partial}}^{p,q}(U)$$

and hence, ∂ -cohomology and Dolbeault cohomology are equal up to conjugation.

Proof:

exercise.

We next upgrade Poincaré's lemma. Let $a = (a_1, \dots, a_n) \in \mathbb{C}^n$, $\epsilon = (\epsilon_1, \dots, \epsilon_n) \in (\mathbb{R}_{>0} \cup \infty)^n$ we denote by

$$\Delta_\epsilon(a) = \{z \in \mathbb{C}^n : |z_i - a_i| < \epsilon_i\}$$

the n -dimensional *polydisk*. For $n = 1$, $a = 0$, $\epsilon = 1$ we set $\Delta = \Delta_1(0)$, the unit disk, and $\Delta^* = \Delta \setminus \{0\}$ the punctured unit disk.

Proposition 1.1.19: $\bar{\partial}$ -Lemma

Let $q \geq 1$, and α be a $\bar{\partial}$ -closed (p, q) -form on a polydisk $\Delta_\epsilon(a)$. Then α is $\bar{\partial}$ -exact; i.e.

$$H_{\bar{\partial}}^{p,q}(\Delta_\epsilon(a)) = 0; q \geq 1.$$

Proof:

exercise in combining complex with diff geo.

Thus, on $U = \Delta_\epsilon$ the p -Dolbeault complex is exact.

1.1.3 Towards Kähler Structure

Every manifold has a Riemann metric [4]. Note every even dimensional manifold has a symplectic structure. If it does, we may ask if it is compatible with the complex structure on some way. We first recall the symplectic structure.

Definition 1.1.20: Symplectic Structure

A *symplectic structure* on a real vector space W is an alternating bilinear form Q on W which is non-degenerate in the sense that the map $h : W \rightarrow W^*$ defined by $h(v) = Q(v, -)$ is an isomorphism.

A symplectic structure on a $2d$ -dimensional smooth manifold M is a closed 2-form $\omega \in \bigwedge^2(M)$ such that

$$\Omega = \frac{\omega^d}{d!}$$

is nowhere vanishing. A *symplectic manifold* is a manifold M endowed with a symplectic structure ω .

Thus, if ω is a symplectic structure on M , at each $p \in M$, the form ω_p defines a symplectic structure Q_p on $T_p(M)$.

Example 1.1.21: Simplest Symplectic Structure

The simplest example of a symplectic manifold is given by $\mathbb{R}^{2d}$ with coordinates $\{x_1, \dots, x_d, y_1, \dots, y_d\}$ and the two-form

$$\omega_0 = \sum_{j=1}^d dx_j \wedge dy_j.$$

The classical Darboux Theorem asserts that, locally, every symplectic manifold is symplectomorphic to $(\mathbb{R}^{2d}, \omega_0)$:

Theorem 1.1.22: Darboux Theorem

Let (M, ω) be a symplectic manifold, then for each $p \in M$ there exists an open neighborhood U and local coordinates $\varphi : U \rightarrow \mathbb{R}^{2d}$ such that $\omega|_U = \varphi^*(\omega_0)$.

Proof:

exercise

Of interest are symplectic structures on a complex manifold M compatible with the complex structure J :

Definition 1.1.23: Hermitian Metric

Let M be a complex manifold and J its complex structure. Then a Riemannian metric g on M is said to be a *Hermitian metric* if and only if for each $p \in M$, the bilinear form g_p on the tangent space $T_p(M)$ is compatible with the complex structure J_p :

$$g_p(Ju, Jv) = g_p(u, v)$$

It is an exercise to check that given a symmetric bilinear form compatible with a complex structure, we can define a J -invariant alternating form. Thus, given a Hermitian metric, we may define a alternative 2-tensor $\omega \in \mathcal{A}^2(M, \mathbb{C})$ via:

$$\omega(X, Y) = g(JX, Y)$$

where if M is complexified then g denote the linear extension on the tangent space. The reader should check that:

$$\omega \in \mathcal{A}^{1,1}(M) \quad \text{and} \quad \omega = \bar{\omega}$$

In coordinates $(U, \{z_1, \dots, z_d\})$, we can write:

$$\omega = \frac{i}{2} \sum_{j,k=1}^d h_{jk} dz_j \wedge d\bar{z}_k \quad \text{where} \quad h_{kj} = \bar{h}_{jk}$$

Thus:

$$\omega \left(\frac{\partial}{\partial x_j}, \frac{\partial}{\partial x_k} \right) = -\text{Im}(h_{jk})$$

and:

$$g \left(\frac{\partial}{\partial x_j}, \frac{\partial}{\partial x_k} \right) = \omega \left(\frac{\partial}{\partial x_j}, \frac{\partial}{\partial y_k} \right) = \text{Re}(h_{jk})$$

Thus g is positive definite if and only if the hermitian matrix $(h_{jk})_{jk}$ is positive definite. By these properties, we can define the Hermitian structure to be a real $(1, 1)$ form ω such that in coordinates as above, $(h_{jk})_{jk}$ is positive definite.

Definition 1.1.24: Kähler Manifold

Let ω be a Hermitian metric on M . Then ω a *Kähler metric* if and only if ω is closed. In this case, the form ω is referred to a *Kähler form*.

A complex manifold is called *Kähler* if it admits a Kähler structure.

Example 1.1.25: Kähler Manifolds

1. on $\mathbb{C}$
2. the other example on p.41

Proposition 1.1.26: Non-trivial Real Cohomology On Compact Kähler

Let M be a compact Kähler d -manifold. Then:

$$\dim H^{2k}(M, \mathbb{R}) > 0$$

for $k = 1, \dots, d$.

Proof:

p.41-42

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